Enterprise Data Management Plan

Apple Fitness+: Improving Customer Engagement and Retention

Management proposal | Kimball CRM dimensional modeling

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Conceptual design; not a description of Apple’s internal systems.

1. The business opportunity

Why a governed engagement data model can improve decisions

BUSINESS QUESTION

Which workout engagement patterns are associated with subscriber retention?

CURRENT-STATE HYPOTHESIS

Relevant data may be distributed across activity, subscription, and content systems. This is a discovery assumption—not a claim about Apple's actual architecture.

SUCCESS MEASURES

- Workouts per active subscriber
- 30-/90-day retention
- Report freshness and reconciliation effort

PROPOSED VALUE

- One trusted set of KPI definitions
- Faster cohort and content analysis
- Less manual reconciliation
- Better evidence for retention experiments

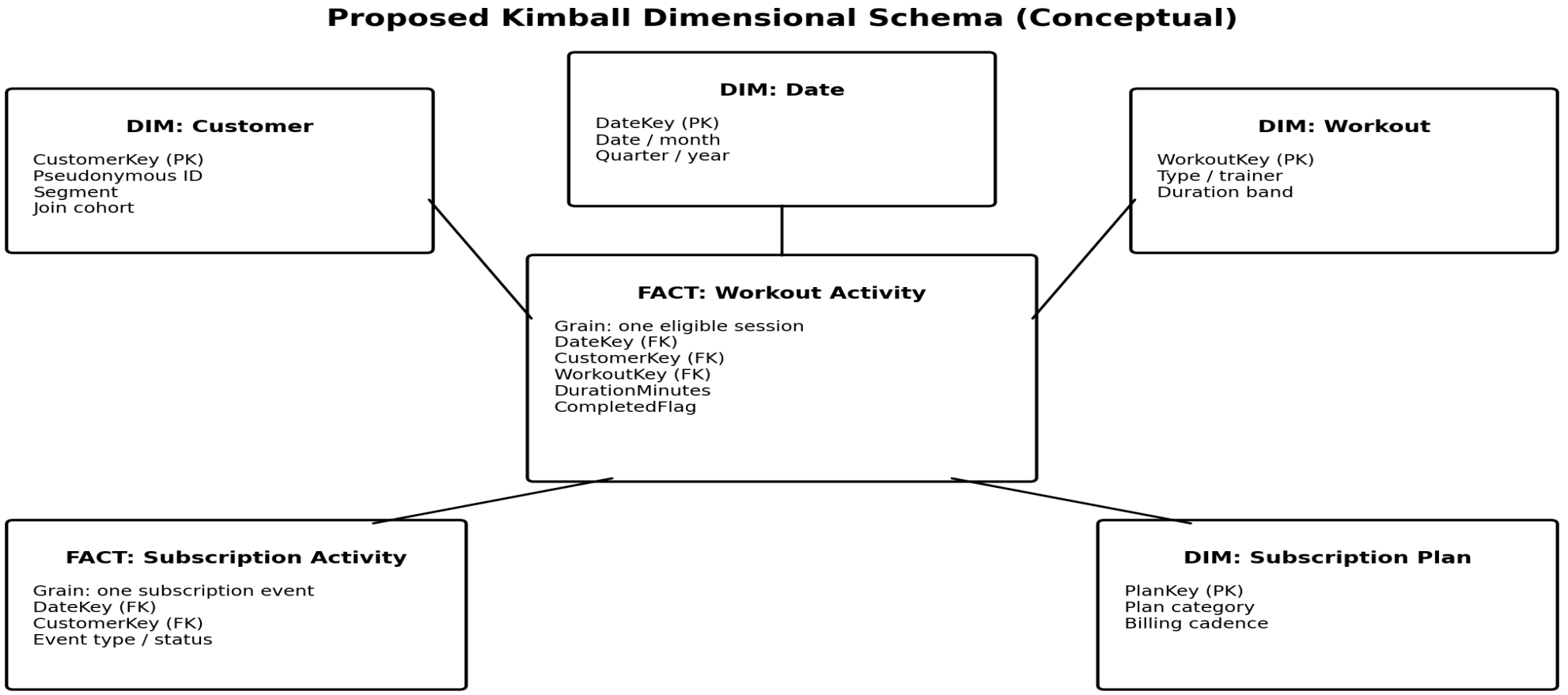
GUARDRAIL

Use approved, minimized, pseudonymous data; avoid unnecessary sensitive health data.



2. Proposed Kimball dimensional schema

Schema = facts, dimensions, keys, relationships, and grain

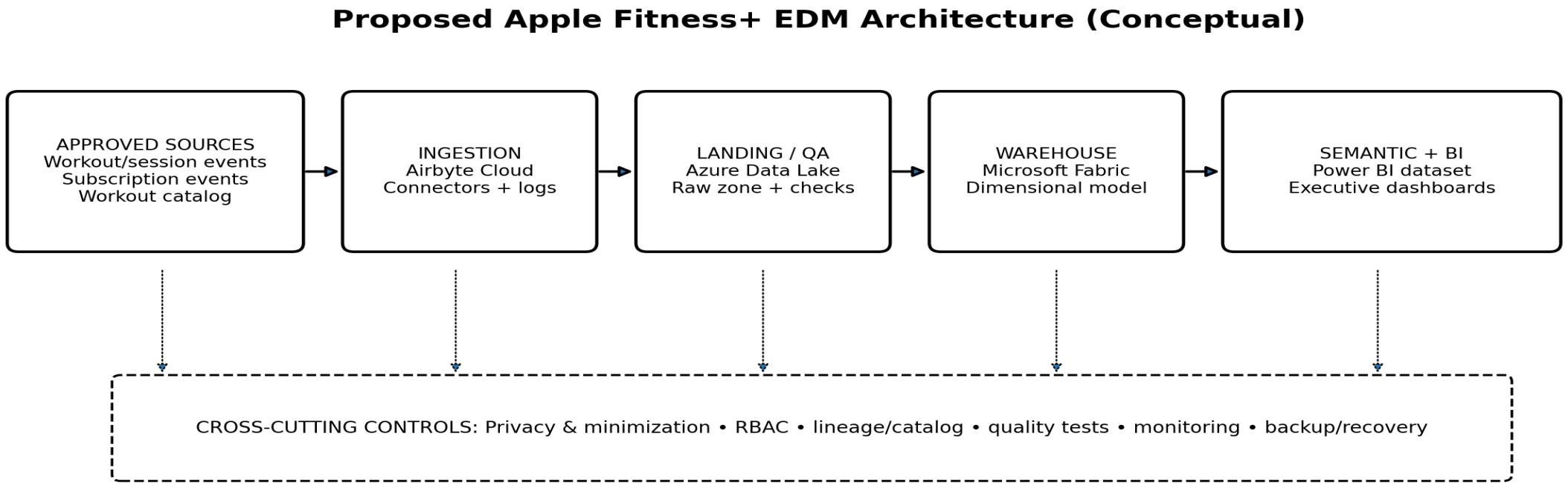


Illustrative schema. Keys, measures, and relationships must be validated against approved source data.

Workout fact grain: one eligible workout session per row. Subscription fact grain: one lifecycle event per row.

3. Architecture: tools and data flow

Architecture = platforms and the movement of data between them



Proposed stack for this academic business case; not a description of Apple's internal systems.

Tools: Airbyte Cloud • Azure Data Lake Storage • Microsoft Fabric Warehouse • Power BI

4. Data journey and control points

Every stage needs an owner, quality check, and recovery path

1 EXTRACT / INGEST

Airbyte moves approved source extracts and logs sync status.

2 LAND / PRESERVE

Raw zone stores timestamped source data with restricted access.

3 VALIDATE / TRANSFORM

Check schemas, required fields, duplicates, keys, and allowed values.

4 MODEL / SERVE

Fabric warehouse builds conformed dimensions and fact tables.

5 ANALYZE

Power BI presents engagement, retention, content mix, and freshness KPIs.

CONTROL THROUGHOUT

RBAC, minimization, lineage, monitoring, retention, audit, backup, and recovery.



5. Staffing and 10-week implementation

Lean pilot assumptions; validate resourcing with source owners

STAFFING (PLANNING ESTIMATES)

- Sponsor / product owner: 0.1 FTE
- Data engineer: 0.5–1.0 FTE
- BI developer / analyst: 0.5 FTE
- Security/privacy reviewer: part-time
- Source-system and data owners: part-time

TIMELINE

Weeks 1–2: discovery and KPI definitions

Weeks 3–4: ingestion and raw zone

Weeks 5–6: dimensional model and transformations

Weeks 7–8: Power BI, reconciliation, UAT

Week 9: parallel run and training

Week 10: launch readiness and handoff



6. Initial cost estimate

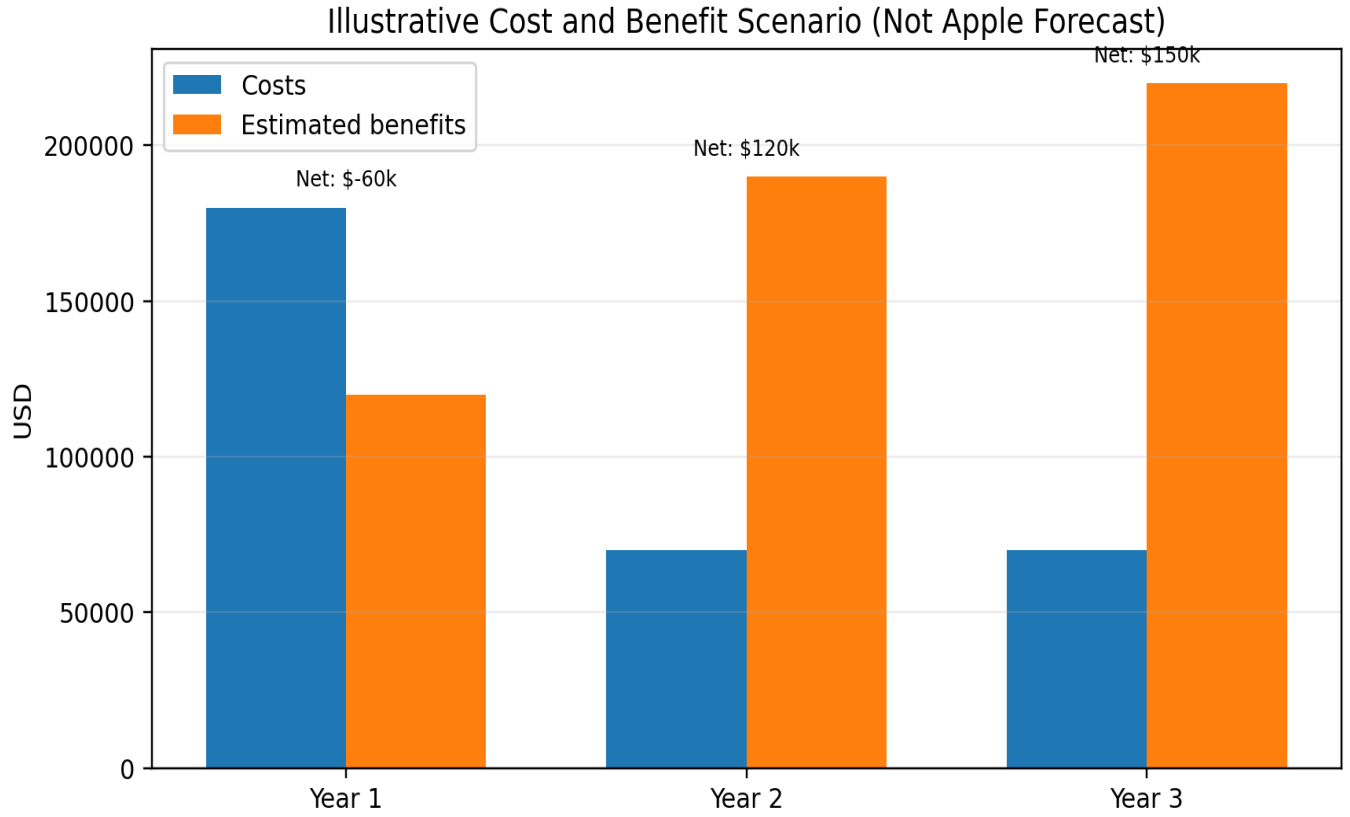
Illustrative USD scenario; not an Apple budget or vendor quote

Labor / implementation	**$120,000**
Cloud, integration, setup	**$20,000**
Training, security, contingency	**$40,000**
Year 1 total	**$180,000**
Annual run cost from Year 2	**$70,000**

Final costs depend on actual data volume, security requirements, licensing, and existing enterprise agreements.

7. Three-year ROI scenario

Benefits are modeled assumptions and must be validated



3-YEAR TOTALS

Benefits: $530,000

Costs: $320,000

Net benefit: $210,000

ROI = (Benefits − Costs) ÷ Costs

Estimated ROI: 65.6%

Year 1 net: −$60,000

Cumulative break-even: during Year 2 if assumptions are achieved.

8. Expected benefits and governance

Measure outcomes; do not confuse correlation with causation

EXPECTED BENEFITS

- Consistent definitions for active engagement
- Faster cohort and content analysis
- Reduced manual reconciliation
- More timely evidence for retention experiments
- Executive visibility into data freshness and quality

ONGOING MAINTENANCE

- Pipeline and freshness alerts
- Schema drift and data-quality checks
- Cost and performance reviews
- Access recertification and audit review
- Retention/deletion enforcement
- Backup, recovery, lineage, and KPI change control



9. Decision requested

Approve a controlled discovery and pilot—not an unqualified enterprise rollout

REQUESTED NEXT STEPS

1. Assign business and data owners.
2. Confirm approved sources, privacy constraints, and baseline KPIs.
3. Validate connectors, Fabric capacity, Power BI licensing, and vendor costs.
4. Run a 10-week pilot with parallel reconciliation and rollback criteria.
5. Expand only if quality, adoption, cost, and value thresholds are met.

The proposal is designed to turn customer activity into governed, decision-ready insight while preserving privacy and financial accountability.



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